


1. what if G is "productive"
 so that $Z = Z(G)$?

if G weren't productive

compared to G unproductive

① if $G \uparrow = T \uparrow$

negative income effect

$U \downarrow C \downarrow L \downarrow N \uparrow$

$Y \uparrow$

$MPN \downarrow W \downarrow$

② $Z \uparrow$ as $Z(G)$

positive income effect

$U \uparrow C \uparrow L \uparrow N \downarrow$

$Y \downarrow$

$MPN \uparrow W \uparrow$

add if $Z \downarrow$

effect but $Y \uparrow$

and substitution effect

$MPN \uparrow (W \uparrow)$

$C \uparrow L \downarrow N \uparrow$

$Y \uparrow$

$MPN \downarrow$

offset

$W \downarrow$

or net $W \uparrow$

}

the fall in w is the result of

$W \uparrow$

$Y \uparrow$

$C \uparrow$

$U \uparrow$

$W \uparrow$

$\Delta L?$

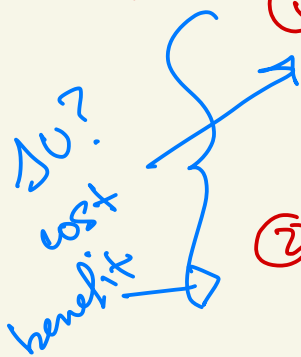
$\Delta N?$

Compare

$G \uparrow$ without $z = z(G)$

to $G \uparrow$ with $z = z(G)$

graphically



① $G \uparrow$

$\Rightarrow \tau \uparrow$? PPF shifts $\downarrow$
no Δ slope
(pure income effect)

② add $z = z(G)$

slope = MPN and $z \uparrow$

$\Rightarrow$ PPF becomes steeper and shifts
up/out $\Rightarrow l^{\max}$ on the horizontal
axis is higher and $c^{\max}$ higher

* Better to have a benefit!

But ① could still be bigger
than ②

Eg2 We assume in ch5

$T = G$ rather
 achieves efficiency than $t \cdot w \cdot N = G$
 market eqm

In ch5

$$MRS = MRT = MPN$$

like ch4

$$\frac{MU(L)}{MU(C)} = w = MPN$$

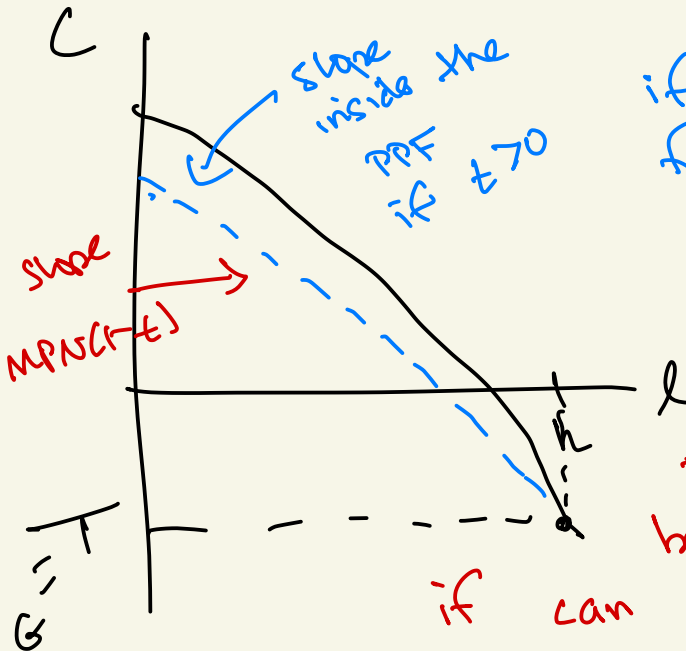
earned by households

if we use "t"
 from ch4:

$$MRS = w(1-t)$$

MRT

$< MPN$



* Inefficient to
 be inside PPF

if can switch from t
 to T without hurting
 anyone.

Ex # 3

1. Assume $U = a \ln c + b \ln l$ } "log linear"

In eqm $\frac{MU(l)}{MU(c)} = w(l-t) = MRT$

$$\left\{ \begin{array}{l} \frac{b/l}{a/c} = \frac{b}{a} \frac{c}{l} = w(l-t) \\ = \frac{b}{a} \frac{(h-l)w(l-t)}{l} = \cancel{w(l-t)} \end{array} \right.$$

$\Rightarrow l \neq f(w(l-t))$
so N isn't either

$t \uparrow$ but
no ΔN
Revenue =
 $t \cdot \bar{w} \cdot \bar{N}$
 $\uparrow$
if $t \uparrow$ Rev $\uparrow$
in ch 4

2. Assume $U = a c + b \ln l$ } "quasi linear"

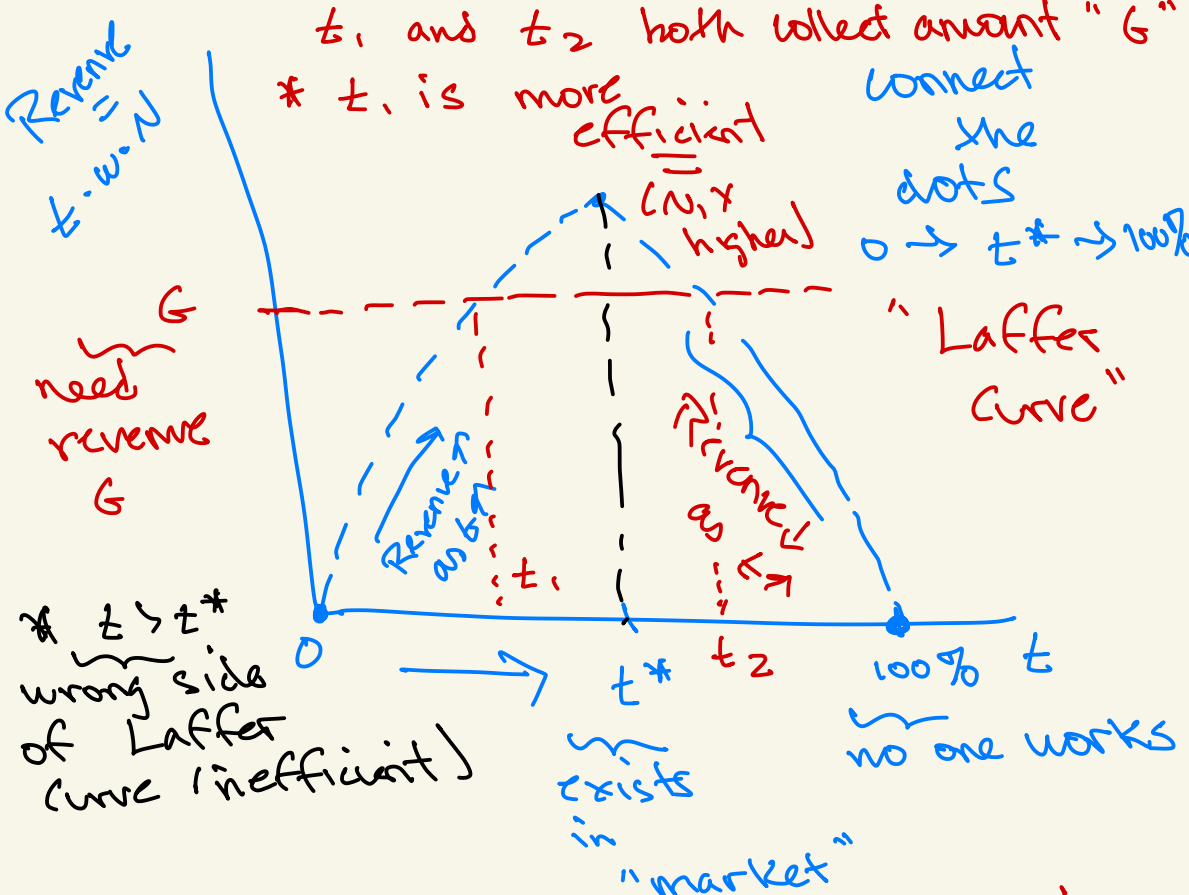
In eqm $\frac{MU(l)}{MU(c)} = \frac{b/l}{a} = w(l-t)$

Revenue
 $t \cdot \bar{w} \cdot \bar{N}$
 $\uparrow \quad \downarrow$

$$\frac{b}{a} \frac{1}{w(l-t)} = l$$

as $t \uparrow$ $l \uparrow$ $N \downarrow$

ambiguous
in ch 4 } if $t \uparrow$ Revenue might $\downarrow$



* Only Laffer curve even in theory if $t \uparrow$ cause $N \downarrow$

in data only in short run?

substitution effects dominate income effects